Richard Harnisch

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Education

MSc Artificial Intelligence September 2026–2028
University of Amsterdam

BSc Artificial Intelligence, cum laude, GPA 8.4/10 2022–2026
University of Groningen

- Thesis: Interpolation Threshold and Double Descent in RL. Tracked Fisher Information Matrix trace across over-parameterized agents; found local curvature can decouple from the generalization gap under policy-induced non-stationarity.
- Formal math/CS foundations: Calculus for Artificial Intelligence; Linear Algebra and Multivariable Calculus; Statistics; Analysis; Probability Theory; Probability and Measure; Linear Algebra 2; Partial Differential Equations; Imperative Programming; Algorithms and Data Structures; Object-Oriented Programming.
- Advanced AI/ML: Applied Machine Learning, Reinforcement Learning, Uncertainty in ML, Machine Translation (9.5 each); Neural Networks, Natural Language Processing (9 each); Advanced Logic (8.5).

International Baccalaureate; German Abitur 1.0 2020–2022
Dresden International School 42/45 points
Higher Level Mathematics, Physics, Economics; maximum marks in Mathematics, Physics, German, Spanish. Extended Essay on public-key cryptography including Euler's totient function, Chinese remainder theorem & vulnerabilities.

Research and ML Experience

Research Assistant: Data Center Optimization 2026–present
Multiscale Networked Systems Group Amsterdam

- Working at the University of Amsterdam under Dr. Shashikant Ilager on Data Center design, targeting publication.
- Optimizing electricity use and carbon intensity while keeping performance within threshold values for both inference and training workloads. Exploring gradient descent on data center component specs in simulations and agent-driven design.

Data Science Intern: Electricity Price Forecasting 2025–2026
Nederlandse Gasunie Groningen/Remote

- Developed a day-ahead electricity price forecasting system using SARIMA, LSTMs, and Transformers in 5-person team.
- Designed and deployed distribution-shift monitoring over ENTSO-E price data and Dutch grid-production data; system alerts operators and triggers retraining under covariate drift; worked directly with Chief Data Scientist.
- Conducted feature engineering and data management for real-world forecasting in a grid undergoing renewable- and geopolitics-driven distribution shifts including alerting operator upon distribution shift to retrain model on new data.

Bachelor's Thesis: Double Descent in Reinforcement Learning 2025–2026
University of Groningen Supervisor: Dr. Matthia Sabatelli

- Investigated the interpolation threshold and Double Descent phenomenon in over-parameterized Reinforcement Learning agents under non-stationary data distributions created by randomized environments and adaptive policy.
- Tracked the Fisher Information Matrix (FIM) trace as a generalization metric across constrained policy optimizers (TRPO) at varying model capacity and training durations. Gained experience with high-performance computing systems.
- Found empirical evidence that FIM decorrelates from the generalization gap under policy-induced non-stationarity: constrained optimizers exhibit expected FIM spike near the interpolation threshold, but the curvature signal does not predict generalization gap evolution, challenging the applicability of classical information-geometric bounds in RL.
- Raises open questions about the role of implicit regularization and optimizer geometry in non-stationary generalization.

Machine Learning Engineer and Software Developer 2024–2026
descript GmbH Dresden/Remote

- Independently led industrial ML research collaboration with TU Dresden on textile defect detection; defined project methodology to maximize learning from limited data and built Django interface for uncertainty-prioritzed expert labelling.
- Engineered production features from client requirements, including a scheduling system for hospitality room management.

Teaching Assistant: Introduction to Machine Learning 2025–2026
University of Groningen

- Led lab sessions and graded assignments for 100 second-year AI students, communicated ML concepts to broad audience.
- Topics: ML paradigms, model selection, regularization, evaluation metrics, data ingestion, processing and augmentation.

Selected Projects

Hackathon Winner. With two teammates, won AI Hackathon Haarlem 2026 with agentic orchestration system that ingests house style based on example `pptx` files and delivers arbitrary editable `pptx` decks including speaker notes & handouts.
Study Association Board. As board of 1,000+ member AI/CS study association at University of Groningen organized AI symposium with academic & industry speakers; ~150 attendees. Organized international excursions to Lisbon & Thessaloniki.
Mars Rover Competition. Authored technical report for European Space Agency rover competition and presented with MakerCie team; placed 1st overall among 33 teams and received maximum points for presentation.

Languages: English (TOEFL iBT 6/6, C2) and German native; Dutch and Spanish conversational.